

CLEARFIELD-LAWRENCE TOWNSHIP, PA, USA

WMO: 725119

Lat: 41.047N

Lon: 78.412W

Elev: 462

StdP: 95.90

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54792

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.9	-14.2	-21.9	0.6	-14.9	-19.4	0.7	-13.1	10.3	-3.5	9.2	-3.3	3.1	240	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	30.8	21.5	29.3	20.7	28.0	20.1	23.1	27.8	22.3	26.6	21.6	25.6	3.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	17.5	25.1	21.1	16.7	24.3	20.3	15.8	23.5	70.9	27.9	67.6	26.5	64.9	25.6	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.7	6.9	-21.2	33.3	3.5	2.0	-23.7	34.7	-25.7	35.9	-27.7	37.1	-30.2	38.6		
(5)			-21.5	24.6	3.4	0.9	-24.0	25.3	-26.0	25.8	-27.9	26.3	-30.4	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-4.1	-2.9	2.4	9.1	14.9	18.8	21.3	20.4	16.8	10.5	4.4	-0.8
	DBStd	10.02	6.13	5.48	5.73	5.15	4.59	3.23	2.80	2.77	3.74	4.75	4.76	4.93
	HDD10.0	1700	438	362	246	77	10	0	0	0	1	53	178	335
	HDD18.3	3572	696	594	494	279	126	34	6	12	72	248	419	592
	CDD10.0	1445	1	1	10	50	162	264	351	323	206	68	9	1
	CDD18.3	276	0	0	0	2	18	48	99	77	27	4	0	0
	CDH23.3	2332	0	0	2	47	211	398	810	593	239	31	1	0
	CDH26.7	604	0	0	0	7	42	92	242	156	61	5	0	0
Wind	WSAvg	2.8	3.4	3.4	3.3	3.3	2.7	2.4	2.2	2.0	2.1	2.7	2.9	3.2
Precipitation	PrecAvg	1067	70	69	90	86	109	113	105	90	95	75	88	79
	PrecMax	1293	152	143	166	161	219	278	258	171	220	174	233	126
	PrecMin	777	21	14	24	19	20	11	38	20	23	9	22	32
	PrecStd	127	37	32	40	37	46	61	44	38	43	38	44	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	14.8	22.1	27.6	29.9	31.0	33.1	32.0	31.1	26.7	20.1	15.1
		MCWB	10.6	10.0	13.7	17.0	19.9	21.5	22.4	22.0	21.3	19.6	13.7	12.7
	2%	DB	10.6	10.3	18.2	24.4	28.0	29.2	30.9	30.0	28.5	23.4	17.1	11.8
		MCWB	8.3	6.5	11.3	14.3	18.4	20.7	21.6	21.3	20.0	17.2	12.0	9.6
	5%	DB	7.1	7.4	14.8	21.4	26.1	27.5	29.2	28.4	26.2	21.1	14.7	8.6
		MCWB	5.3	4.5	9.4	12.6	17.4	20.0	21.0	20.7	19.2	16.2	10.6	6.4
	10%	DB	4.3	4.7	11.9	18.5	23.6	25.7	27.7	26.9	23.8	18.5	12.4	6.1
		MCWB	2.5	2.1	8.1	11.1	16.4	18.9	20.5	19.9	18.0	14.7	9.1	4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	11.4	15.0	18.1	21.3	22.9	24.6	24.2	23.5	20.9	15.6	13.1
		MCDB	14.1	13.3	20.2	24.4	27.4	27.9	29.8	28.9	28.2	25.0	18.2	14.6
	2%	WB	8.3	7.5	12.9	16.0	20.1	21.9	23.3	22.9	21.7	19.1	13.3	10.0
		MCDB	9.7	9.9	16.6	21.3	25.3	26.9	28.3	27.2	25.6	21.9	15.8	11.3
	5%	WB	5.5	4.7	10.7	14.4	19.1	21.2	22.4	22.0	20.5	17.2	11.4	6.9
		MCDB	7.2	7.0	13.9	19.2	23.8	25.7	26.9	25.5	24.1	19.7	13.9	8.3
	10%	WB	2.7	2.3	7.9	12.6	17.8	20.3	21.6	21.2	19.4	15.1	9.5	4.5
		MCDB	4.1	4.6	11.2	16.9	21.9	24.1	25.6	24.8	22.4	18.3	12.0	6.0
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.8	10.3	12.3	12.2	11.4	11.7	11.6	11.7	10.6	9.4	7.1
		MCDBR	11.1	12.5	15.4	17.5	15.1	13.6	13.3	13.7	14.5	14.4	13.8	10.7
		MCWBR	9.3	9.5	9.5	9.0	7.1	6.5	5.6	5.9	6.9	8.1	9.2	8.8
	5% WB	MCDBR	10.7	11.4	13.6	14.3	12.3	11.4	11.3	11.0	11.4	11.6	11.5	9.9
		MCWBR	9.4	9.4	9.6	8.6	6.2	5.9	5.3	5.3	5.8	7.1	8.8	8.9
Clear-Sky Solar Irradiance	taub	0.310	0.329	0.351	0.374	0.409	0.457	0.462	0.455	0.375	0.366	0.323	0.301	
	taud	2.387	2.308	2.350	2.336	2.257	2.145	2.142	2.153	2.363	2.381	2.480	2.467	
	Ebn at Noon	841	875	897	898	869	825	815	809	861	824	823	826	
	Edh at Noon	82	103	112	121	135	151	150	144	109	96	75	71	
All-Sky Solar Radiation	RadAvg	1.53	2.35	3.34	4.47	5.08	5.56	5.65	5.05	4.10	2.73	1.83	1.23	
	RadStd	0.12	0.28	0.33	0.45	0.51	0.47	0.41	0.30	0.49	0.30	0.23	0.14	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (34 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page